Lecture Notes

Lecture 8

Orthonormal Bases, Gram–Schmidt, and Riesz

Lecture 7 ended with a finite-dimensional question: can we find a basis of mutually orthogonal unit vectors, and expand every vector as a finite sum in it? Over $\mathbb{R}$ or $\mathbb{C}$, this lecture answers yes. The *Gram–Schmidt procedure* manufactures an *orthonormal basis* from any basis, and in such a basis everything becomes simple: coordinates are just inner products, $v = \sum_k \langle e_k, v\rangle\, e_k$; the norm is the Pythagorean sum $\|v\|^2 = \sum_k |\langle e_k, v\rangle|^2$; and the identity resolves as $\sum_k |e_k\rangle\langle e_k| = I$. We then use orthonormality to solve the *best-approximation* problem (orthogonal projection finds the closest point of a subspace) and to prove the *Riesz representation theorem*, which says every bra is the partner of a unique ket. These are the daily tools of quantum mechanics: expanding a state in an energy eigenbasis, computing amplitudes, and projecting onto measurement outcomes.

Learning objectives

By the end of this lecture you will be able to:

- Expand vectors in an orthonormal basis, with coordinates as inner products.
- Carry out the Gram–Schmidt procedure to build an orthonormal basis.
- Compute orthogonal projections and solve best-approximation / least-squares problems.
- State the Riesz representation theorem and link it to the bra–ket duality.

1 Orthonormal bases

Definition 8.1 (Orthonormal list and basis). A list $e_1, \ldots, e_m$ is *orthonormal* if $\langle e_j, e_k\rangle = \delta_{jk}$ (each is a unit vector, distinct ones orthogonal). An orthonormal list that is also a basis is an *orthonormal basis* (ONB).

Lemma 8.2 (Orthonormal lists are independent). Every orthonormal list is linearly independent.

Proof. If $\sum_k c_k e_k = 0$, pairing with e_j gives $0 = \langle e_j, \sum_k c_k e_k\rangle = \sum_k c_k \delta_{jk} = c_j$. So every coefficient vanishes. ∎

The point of an ONB is that coordinates require no linear system—they are inner products.

Theorem 8.3 (Expansion in an orthonormal basis). If $e_1, \ldots, e_n$ is an orthonormal basis of V, then for every $v \in V$

$$v = \sum_{k=1}^{n} \langle e_k, v\rangle\, e_k, \qquad \|v\|^2 = \sum_{k=1}^{n} |\langle e_k, v\rangle|^2.$$

The second identity is *Parseval's relation.*

Proof. Write $v = \sum_k c_k e_k$ (possible since the e_k are a basis). Pairing with e_j and using orthonormality, $\langle e_j, v\rangle = \sum_k c_k \delta_{jk} = c_j$, which gives the first formula. Then $\|v\|^2 = \langle v, v\rangle = \sum_{j,k} \overline{c_j} c_k \langle e_j, e_k\rangle = \sum_k |c_k|^2$, the second. ■

Corollary 8.4 (Inner products from coordinates). In an orthonormal basis, $\langle u, v\rangle = \sum_k \overline{\langle e_k, u\rangle}\,\langle e_k, v\rangle$. Once vectors are expanded in an ONB, every inner product is the standard sum $\sum_k \overline{a_k} b_k$ of their coordinates—which is why $\mathbb{C}^n$ with its standard product is the universal model: any n-dimensional complex inner product space is isometric to it.

Proof. Expand $u = \sum_j \langle e_j, u\rangle e_j$ and $v = \sum_k \langle e_k, v\rangle e_k$; then

$$\langle u, v\rangle = \sum_{j,k} \overline{\langle e_j, u\rangle}\langle e_k, v\rangle\langle e_j, e_k\rangle = \sum_k \overline{\langle e_k, u\rangle}\langle e_k, v\rangle.$$ ■

Remark 8.5 (Completeness relation). In Dirac notation the expansion reads

$$|v\rangle = \sum_k |e_k\rangle\langle e_k, v\rangle = \Big(\sum_k |e_k\rangle\langle e_k|\Big)|v\rangle,$$

so an orthonormal basis satisfies the *completeness relation* $\sum_k |e_k\rangle\langle e_k| = I$. Inserting this resolution of the identity between symbols is the basic move of quantum-mechanical calculation.

Example 8.6 (Finite orthonormal-basis examples). The standard basis of $\mathbb{C}^n$ is orthonormal. So is the qubit pair $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ of Lecture 7.

Bridge (finite Fourier modes). On the space of finite Fourier series, the functions $\frac{1}{\sqrt{2\pi}}$ and, for $k \ge 1$, $\frac{1}{\sqrt{\pi}}\cos kx$, $\frac{1}{\sqrt{\pi}}\sin kx$ are orthonormal for the inner product $\langle f, g\rangle = \int_{-\pi}^{\pi} \overline{f} g\, dx$—the basis behind Fourier analysis. ◁

Example 8.7 (Coordinates as inner products). Expand $v = (3, 4) \in \mathbb{R}^2$ in the ONB $e_1 = \frac{1}{\sqrt{2}}(1, 1)$, $e_2 = \frac{1}{\sqrt{2}}(1, -1)$. The coordinates are $\langle e_1, v\rangle = \frac{7}{\sqrt{2}}$ and $\langle e_2, v\rangle = -\frac{1}{\sqrt{2}}$, so $v = \frac{7}{\sqrt{2}}e_1 - \frac{1}{\sqrt{2}}e_2$, and Parseval checks: $\frac{49}{2} + \frac{1}{2} = 25 = \|v\|^2$. No linear system was solved. ◁

Example 8.8 (Bridge: a Fourier coefficient). In the Fourier basis above, the coordinate of a signal f along $\frac{1}{\sqrt{\pi}}\sin kx$ is $\frac{1}{\sqrt{\pi}}\int_{-\pi}^{\pi} \sin(kx) f(x)\, dx$, the kth Fourier sine coefficient. Parseval then equates the “energy” $\|f\|^2$ with the sum of squared coefficients—the foundation of spectral analysis. ◁

Example 8.9 (A complex orthonormal basis). On $\mathbb{C}^2$ the vectors $e_1 = \frac{1}{\sqrt{5}}(2, i)$ and $e_2 = \frac{1}{\sqrt{5}}(1, -2i)$ are orthonormal: $\|e_1\|^2 = \frac{1}{5}(4+1) = 1$, $\|e_2\|^2 = \frac{1}{5}(1+4) = 1$, and $\langle e_1, e_2\rangle = \frac{1}{5}\big(\overline{2}\cdot 1 + \overline{i}\cdot(-2i)\big) = \frac{1}{5}(2-2) = 0$. The conjugation in the first slot is exactly what produces the cancellation. ◁

2 The Gram–Schmidt procedure

Theorem 8.10 (Gram–Schmidt). Every finite-dimensional inner product space has an orthonormal basis. Concretely, from any basis $v_1, \dots, v_n$ one builds an orthonormal basis $e_1, \dots, e_n$ with $\mathrm{span}(e_1, \dots, e_k) = \mathrm{span}(v_1, \dots, v_k)$ for every k, by

$$e_1 = \frac{v_1}{\|v_1\|}, \qquad e_k = \frac{w_k}{\|w_k\|}, \qquad w_k = v_k - \sum_{j<k} \langle e_j, v_k \rangle \, e_j.$$

Proof. We induct on k, assuming $e_1, \dots, e_{k-1}$ orthonormal with the right span. The numerator $w_k = v_k - \sum_{j<k} \langle e_j, v_k \rangle e_j$ is nonzero, else v_k would lie in $\mathrm{span}(v_1, \dots, v_{k-1})$, contradicting independence. For $i < k$,

$$\langle e_i, w_k \rangle = \langle e_i, v_k \rangle - \sum_{j<k} \langle e_j, v_k \rangle \langle e_i, e_j \rangle = \langle e_i, v_k \rangle - \langle e_i, v_k \rangle = 0,$$

so $e_k = w_k / \|w_k\|$ is a unit vector orthogonal to the earlier ones. Since v_k and e_k differ only by a scalar and a combination of $e_1, \dots, e_{k-1}$, the spans agree. After n steps we have n orthonormal vectors, a basis by Lemma 8.2. ■

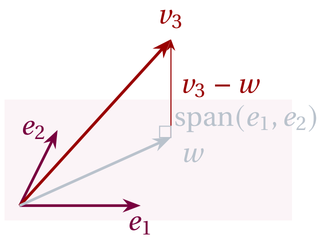

Figure 1: The third Gram–Schmidt step: subtract from v_3 its projection $w = \langle e_1, v_3 \rangle e_1 + \langle e_2, v_3 \rangle e_2$ onto $\mathrm{span}(e_1, e_2)$; the remainder $v_3 - w$ is orthogonal to the plane and is normalized to give e_3.

Example 8.11 (Gram–Schmidt in $\mathbb{R}^3$). Apply the procedure to $v_1 = (1, 1, 0)$, $v_2 = (2, 0, 1)$, $v_3 = (1, -2, -2)$. First $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$. Then

$$v_2 - \langle e_1, v_2 \rangle e_1 = (2, 0, 1) - (1, 1, 0) = (1, -1, 1), \qquad e_2 = \tfrac{1}{\sqrt{3}}(1, -1, 1).$$

Finally $v_3 - \langle e_1, v_3 \rangle e_1 - \langle e_2, v_3 \rangle e_2 = \frac{7}{6}(1, -1, -2)$, so $e_3 = \frac{1}{\sqrt{6}}(1, -1, -2)$. One checks $\langle e_i, e_j \rangle = \delta_{ij}$. ◁

Example 8.12 (Verifying orthonormality). For the output of Example 8.11, the off-diagonal checks are $\langle e_1, e_2 \rangle = \frac{1}{\sqrt{6}}(1 - 1 + 0) = 0$, $\langle e_1, e_3 \rangle = \frac{1}{\sqrt{12}}(1 - 1 + 0) = 0$, and $\langle e_2, e_3 \rangle = \frac{1}{\sqrt{18}}(1 + 1 - 2) = 0$, while each $\|e_i\| = 1$. Confirming $\langle e_i, e_j \rangle = \delta_{ij}$ directly is always the way to validate a Gram–Schmidt computation. ◁

Example 8.13 (Gram–Schmidt over $\mathbb{C}$). Orthonormalize $v_1 = (1, i)$, $v_2 = (1, 0)$ in $\mathbb{C}^2$. First $e_1 = \frac{1}{\sqrt{2}}(1, i)$. Then $\langle e_1, v_2 \rangle = \frac{1}{\sqrt{2}} \overline{1} \cdot 1 = \frac{1}{\sqrt{2}}$, so $v_2 - \langle e_1, v_2 \rangle e_1 = (1, 0) - \frac{1}{2}(1, i) = \frac{1}{2}(1, -i)$, which normalizes to $e_2 = \frac{1}{\sqrt{2}}(1, -i)$. A check gives $\langle e_1, e_2 \rangle = \frac{1}{2}(1 + i^2) = 0$; the conjugations must be tracked carefully over $\mathbb{C}$. ◁

Example 8.14 (Bridge: Legendre polynomials). On $\mathcal{P}_2$ with $\langle p, q \rangle = \int_{-1}^{1} \overline{p} \, q \, dx$, orthonormalize $1, x, x^2$. Since $\int_{-1}^{1} 1 \, dx = 2$, $p_0 = \frac{1}{\sqrt{2}}$. As $\langle p_0, x \rangle = 0$, x is already orthogonal to

p_0; normalizing, $p_1 = \sqrt{\frac{3}{2}}\,x$. For the quadratic, $x^2 - \langle p_0, x^2\rangle p_0 - \langle p_1, x^2\rangle p_1 = x^2 - \frac{1}{3}$, and normalizing gives

$$p_2 = \sqrt{\tfrac{5}{8}}\,(3x^2 - 1).$$

These are the first *Legendre polynomials*, ubiquitous in electrostatics and the quantum theory of angular momentum. ◁

Remark 8.15 (Gram determinant and volume). In a real inner-product space, the determinant of the Gram matrix $G_{jk} = \langle v_j, v_k\rangle$ equals the squared volume of the parallelepiped spanned by $v_1, \dots, v_m$, and it is nonzero exactly when the vectors are independent (Lecture 7). Gram–Schmidt computes the same content step by step: the norms $\|w_k\|$ are the successive "heights" whose product is that volume.

Example 8.16 (Orthonormal basis of a plane). The plane $U = \{(x, y, z) : x - y - 2z = 0\} \subset \mathbb{R}^3$ contains the independent vectors $v_1 = (2, 0, 1)$ and $v_2 = (3, 1, 1)$. Gram–Schmidt gives $e_1 = \frac{1}{\sqrt{5}}(2, 0, 1)$, and from $v_2 - \langle e_1, v_2\rangle e_1 = (3, 1, 1) - \frac{7}{5}(2, 0, 1) = \frac{1}{5}(1, 5, -2)$, $e_2 = \frac{1}{\sqrt{30}}(1, 5, -2)$. These orthonormal vectors span U. ◁

> **Corollary 8.17 (Extending to an orthonormal basis).** Every orthonormal list in a finite-dimensional V extends to an orthonormal basis of V.

Proof. Extend the orthonormal list to a basis (Lecture 2), then run Gram–Schmidt; the already-orthonormal vectors are unchanged, and the rest become an orthonormal completion. ■

Remark 8.18 (Bridge: the QR factorization). Collecting the relations $v_k = \sum_{j \le k} r_{jk} e_j$ into matrices says $A = QR$: any matrix with independent columns factors as a matrix Q with orthonormal columns times an upper-triangular R. For dense numerical least squares, Householder QR is a standard robust choice; Gram–Schmidt remains the transparent geometric construction.

Remark 8.19 (Orthonormal bases are far from unique). For a fixed ordered independent input, the displayed normalization $e_k = w_k/\|w_k\|$ gives a unique ordered output. Changing the input order changes that output; deliberately changing signs or complex phases afterwards produces other ONBs. Every nonzero complex space, and every real space of dimension at least two, has infinitely many ONBs; $\mathbb{R}$ has only (1) and (-1), while the zero space has one. Any two ONBs are related by a *unitary* change of basis—the inner-product-preserving operators of Lecture 9.

Remark 8.20 (Bridge: concise numerical caution). In floating-point arithmetic the classical formula can lose orthogonality through accumulated rounding. *Modified* Gram–Schmidt, which subtracts each projection as soon as it is computed, is usually more stable than the classical form; Householder QR is a standard robust choice for dense least-squares work. The mathematics is identical, the numerical behaviour is not.

3 Orthogonal projection and best approximation

Let U be a subspace of V with orthonormal basis $e_1, \dots, e_m$. The *orthogonal projection* onto U is

$$P_U v = \sum_{k=1}^{m} \langle e_k, v\rangle\, e_k.$$

Theorem 8.21 (Orthogonal decomposition). For each $v \in V$, $v - P_U v \in U^\perp$, and $V = U \oplus U^\perp$. Thus P_U is the projection onto U along $U^\perp$.

Proof. For each i, $\langle e_i, v - P_U v\rangle = \langle e_i, v\rangle - \sum_k \langle e_k, v\rangle\langle e_i, e_k\rangle = \langle e_i, v\rangle - \langle e_i, v\rangle = 0$, so $v - P_U v$ is orthogonal to every e_i and hence to U. Then $v = P_U v + (v - P_U v)$ exhibits v as a sum of a vector in U and one in $U^\perp$; since $U \cap U^\perp = \{0\}$ (a vector orthogonal to itself is 0), the sum is direct. ■

Proposition 8.22 (Dimension of the orthogonal complement). For a subspace U of a finite-dimensional V, $\dim U + \dim U^\perp = \dim V$, and $(U^\perp)^\perp = U$.

Proof. Take an orthonormal basis $e_1, \dots, e_m$ of U and extend it to an orthonormal basis $e_1, \dots, e_n$ of V (Corollary 8.17). The extra vectors $e_{m+1}, \dots, e_n$ lie in $U^\perp$ and span it—any $v \in U^\perp$ has zero coordinates along $e_1, \dots, e_m$—so $\dim U^\perp = n - m$. For the second claim, $U \subseteq (U^\perp)^\perp$ straight from the definition: every $u \in U$ is orthogonal to every vector of $U^\perp$. Applying the dimension identity to $U^\perp$ gives $\dim(U^\perp)^\perp = n - (n - m) = m = \dim U$, and a subspace contained in another of the same finite dimension equals it, so $(U^\perp)^\perp = U$. ■

Example 8.23 (Computing an orthogonal complement). For $U = \operatorname{span}\big((1, 2, 3, -4)\big) \subset \mathbb{R}^4$, the complement $U^\perp$ is the solution set of $\langle (1, 2, 3, -4), v\rangle = v_1 + 2v_2 + 3v_3 - 4v_4 = 0$, a three-dimensional hyperplane—consistent with $\dim U + \dim U^\perp = 1 + 3 = 4$. Reading off the free-variable solutions gives a basis $(-2, 1, 0, 0)$, $(-3, 0, 1, 0)$, $(4, 0, 0, 1)$ of $U^\perp$. ◁

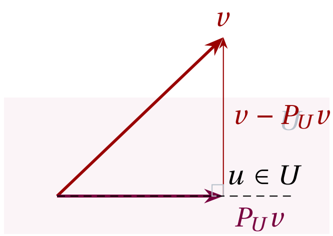

Figure 2: Best approximation: of all points $u \in U$, the orthogonal projection $P_U v$ is closest to v, because the error $v - P_U v$ is perpendicular to U.

Theorem 8.24 (Best approximation). For every $u \in U$, $\|v - P_U v\| \le \|v - u\|$, with equality only when $u = P_U v$.

Proof. Write $v - u = (v - P_U v) + (P_U v - u)$. The first piece lies in $U^\perp$ and the second in U, so they are orthogonal; by the Pythagorean theorem

$$\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \ge \|v - P_U v\|^2,$$

with equality iff $\|P_U v - u\| = 0$, i.e. $u = P_U v$. ■

Example 8.25 (A projection as a matrix). Onto the line $\operatorname{span}(e)$ with $e = \frac{1}{\sqrt{2}}(1, 1)$, the projection operator is the outer product $|e\rangle\langle e| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. It satisfies $P^2 = P$ and $P^\dagger = P$—the algebraic signature of an orthogonal projection that Lecture 9 will make precise. ◁

Corollary 8.26 (Bessel's inequality). For any orthonormal list $e_1, \ldots, e_m$ and any v, $\sum_{k=1}^{m} |\langle e_k, v\rangle|^2 \le \|v\|^2$.

Proof. With $U = \operatorname{span}(e_1, \ldots, e_m)$, Parseval on U gives $\|P_U v\|^2 = \sum_k |\langle e_k, v\rangle|^2$, and $\|P_U v\| \le \|v\|$ since $v - P_U v \perp P_U v$ (Pythagoras again). ■

Example 8.27 (Bessel in action). For $v = (1, 2, 2) \in \mathbb{R}^3$ and the single unit vector $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $|\langle e_1, v\rangle|^2 = \frac{1}{2}(1+2)^2 = \frac{9}{2}$, while $\|v\|^2 = 9$. Bessel's inequality $\frac{9}{2} \le 9$ holds, and the gap $9 - \frac{9}{2} = \frac{9}{2}$ is exactly the squared distance from v to the line—the part of v the single vector cannot capture. ◁

Example 8.28 (Projection onto a line and least squares). Onto the line through a unit vector e, the projection is $Pv = \langle e, v\rangle\, e$; for a general nonzero a, $Pv = \dfrac{\langle a, v\rangle}{\|a\|^2}\, a$. This is the elementary projection formula used throughout the core.

Bridge (normal equations and the adjoint). Least-squares fitting is projection in disguise: to satisfy an overdetermined system $Ax = b$ as nearly as possible, project b onto range A. Best approximation says the residual is orthogonal to the range, $\langle Ay, b - Ax\rangle = 0$ for every y; rewriting the left side as $\langle y, A^\dagger(b - Ax)\rangle$—the defining property of the adjoint, stated in the Riesz bridge at the end of this lecture—gives the *normal equations*

$$A^\dagger A x = A^\dagger b,$$

whose solution minimizes $\|Ax - b\|$. The retained line-fit example below is core; this operator-level derivation is a forward bridge to Lecture 9. ◁

Example 8.29 (Least-squares line fit). To fit $y = c_0 + c_1 x$ to the points $(0, 1), (1, 3), (2, 4)$, set up the overdetermined system $Ac = y$ with

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}.$$

The normal equations $A^\dagger A c = A^\dagger y$ become $\left(\begin{smallmatrix} 3 & 3 \\ 3 & 5 \end{smallmatrix}\right) c = \left(\begin{smallmatrix} 8 \\ 11 \end{smallmatrix}\right)$, with solution $c = (c_0, c_1) = \left(\frac{7}{6}, \frac{3}{2}\right)$. The line $y = \frac{7}{6} + \frac{3}{2}x$ minimizes the total squared vertical error—precisely the orthogonal projection of y onto range A. ◁

Example 8.30 (Physics bridge: a projective measurement). A spin in state $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$, measured in the $\{|0\rangle, |1\rangle\}$ basis, yields $|0\rangle$ with probability $|\langle 0, \psi\rangle|^2 = \frac{1}{2}$ and $|1\rangle$ with probability $\frac{1}{2}$. The post-measurement state is the normalized projection onto the observed eigenspace—here $|0\rangle$ or $|1\rangle$. This is the ideal projective (Lueders) measurement model: it uses orthogonal projection followed by renormalization. General measurements need not have this update rule. ◁

Example 8.31 (Optional extension: degree-five sine fit). Projecting $v(x) = \sin x$ onto the degree-≤ 5 polynomials in $C[-\pi, \pi]$ (run Gram–Schmidt on $1, x, \ldots, x^5$, then apply P_U) gives the best mean-square approximation

$$u(x) \approx 0.98786\, x - 0.15527\, x^3 + 0.00564\, x^5,$$

with smaller global mean-square error than the degree-five Taylor polynomial $x - \frac{x^3}{3!} + \frac{x^5}{5!}$. Orthogonal projection, not Taylor expansion, is the right tool for L^2 or mean-square quality; it does not minimize the pointwise worst-case error. For example, on $[0, 1]$ the L^2-best constant for x^2 is $\int_0^1 x^2\, dx = 1/3$, whereas the uniform best constant is $1/2$. ◁

Example 8.32 (Closest point in $\mathbb{R}^4$). To find the point of $U = \operatorname{span}\big((1,1,0,0),(1,1,1,2)\big)$ nearest $b = (1,2,3,4)$, orthonormalize: $e_1 = \frac{1}{\sqrt{2}}(1,1,0,0)$, and since $(1,1,1,2) - \langle e_1,(1,1,1,2)\rangle e_1 = (0,0,1,2)$, $e_2 = \frac{1}{\sqrt{5}}(0,0,1,2)$. Then

$$P_U b = \langle e_1,b\rangle e_1 + \langle e_2,b\rangle e_2 = \tfrac{3}{2}(1,1,0,0) + \tfrac{11}{5}(0,0,1,2) = \big(\tfrac{3}{2},\tfrac{3}{2},\tfrac{11}{5},\tfrac{22}{5}\big),$$

the least-squares closest point. ◁

Example 8.33 (Projection onto a plane). With the orthonormal e_1, e_2 of Example 8.16, the projection onto that plane is $P_U = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|$. Acting on $b = (1,0,0)$,

$$P_U b = \langle e_1,b\rangle e_1 + \langle e_2,b\rangle e_2 = \tfrac{2}{5}(2,0,1) + \tfrac{1}{30}(1,5,-2) = \tfrac{1}{6}(5,1,2),$$

and the residual $b - P_U b = \frac{1}{6}(1,-1,-2)$ is parallel to the plane's normal $(1,-1,-2)$, as the orthogonal decomposition demands. ◁

4 The Riesz representation theorem

A *linear functional* on V is a linear map $\varphi\colon V \to \mathbb{F}$, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$. In an inner product space every functional is "take the inner product with a fixed vector."

> **Theorem 8.34 (Riesz representation).** Let V be a finite-dimensional inner-product space over $\mathbb{F}$. For every linear functional $\varphi\colon V \to \mathbb{F}$ there is a unique $w \in V$ with
>
> $$\varphi(v) = \langle w, v\rangle \qquad \text{for all } v \in V.$$

Proof. Fix an orthonormal basis $e_1,\dots,e_n$ and set $w = \sum_k \overline{\varphi(e_k)}\, e_k$. Then for any $v = \sum_k \langle e_k, v\rangle e_k$,

$$\langle w, v\rangle = \sum_k \varphi(e_k)\langle e_k, v\rangle = \varphi\Big(\sum_k \langle e_k, v\rangle e_k\Big) = \varphi(v),$$

using $\langle w, e_k\rangle = \varphi(e_k)$ in the first step and linearity of φ in the second. For uniqueness, if $\langle w,v\rangle = \langle w',v\rangle$ for all v then $\langle w-w', v\rangle = 0$ for all v; taking $v = w - w'$ forces $w = w'$. ■

Remark 8.35 (Bra–ket duality). Riesz says exactly that the map $|w\rangle \mapsto \langle w|$ is a bijection: every *bra* (linear functional φ) is $\langle w|$ for a unique *ket* $|w\rangle$. The bra space is thus an exact (conjugate-linear) copy of the ket space—the rigorous content of Dirac's notation. The conjugation $w = \sum_k \overline{\varphi(e_k)} e_k$ is the same conjugation that makes $|\phi\rangle \mapsto \langle\phi|$ conjugate-linear over $\mathbb{C}$. Over $\mathbb{R}$ this correspondence is linear because conjugation is trivial.

Example 8.36 (A represented functional). On $\mathbb{C}^2$ the functional $\varphi(v) = v_1 + 2iv_2$ is $\langle w, v\rangle$ with $w = (\overline{1}, \overline{2i}) = (1, -2i)$, since $\langle w, v\rangle = \overline{1}\, v_1 + \overline{(-2i)} v_2 = v_1 + 2iv_2$. The conjugation lands the coefficients in the right slot. ◁

Example 8.37 (Bridge: Riesz on a polynomial function space). "Evaluate at 0" is a linear functional on $\mathcal{P}_n$ with the L^2 inner product, so by Riesz it equals $\langle w, \cdot\rangle$ for a fixed polynomial w—a *reproducing kernel*: $p(0) = \langle w, p\rangle$ for all p. Every linear functional on a finite-dimensional inner product space conceals such a representing vector. ◁

Example 8.38 (Bridge: Frobenius Riesz and the trace). On $M_n(\mathbb{C})$ with the Frobenius product, the trace functional $\varphi(A) = \operatorname{tr} A$ is represented by the identity matrix: $\langle I, A\rangle = \operatorname{tr}(I^\dagger A) = \operatorname{tr} A$, so $w = I$. Even "sum the diagonal" is secretly an inner product with a fixed vector. ◁

Remark 8.39 (Bridge: from Riesz to the adjoint). Riesz is the engine of the next lecture. Given $T \in \mathcal{L}(V)$ and a fixed u, the map $v \mapsto \langle u, Tv\rangle$ is a linear functional, so by Riesz it equals $\langle w, v\rangle$ for a unique w; defining $T^\dagger u = w$ produces the *adjoint*, characterized by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$. The mere existence of adjoints is Riesz in disguise.

5 Physics bridge: measurement, expansion, and what comes next

In the finite/discrete ideal-projective model, scalar outcomes are eigenvalues λ, including degenerate ones. Their spectral projections P_λ have mutually orthogonal eigenspaces, and a normalized pure-state representative $|\psi\rangle$ yields outcome λ with probability $\|P_\lambda\psi\|^2$. Parseval expresses normalization as the sum of these probabilities. The Lueders update postulate sends a state with nonzero outcome probability to the normalized projection $P_\lambda|\psi\rangle/\|P_\lambda\psi\|$. General measurements lie outside this core model; continuous-spectrum spectral measures wait until Lecture 14.

Example 8.40 (Physics bridge: probabilities sum to one). If $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}\,|1\rangle$, the outcome probabilities are $|\langle 0, \psi\rangle|^2 = \frac{1}{3}$ and $|\langle 1, \psi\rangle|^2 = \frac{2}{3}$, summing to 1 precisely because Parseval expresses the normalization condition as $\|\psi\|^2 = \frac{1}{3} + \frac{2}{3} = 1$. ◁

Example 8.41 (Physics bridge: a time-independent energy basis). For a finite/discrete, time-independent Hamiltonian with orthonormal energy eigenbasis, $|\psi\rangle = \sum_n \langle E_n, \psi\rangle|E_n\rangle$ and the time-evolved state is $|\psi(t)\rangle = \sum_n e^{-iE_n t/\hbar}\langle E_n, \psi\rangle|E_n\rangle$ (Lecture 6): each coordinate merely rotates by a phase. The probabilities $|\langle E_n, \psi\rangle|^2$ are unchanged in time, expressing conservation of energy. ◁

Remark 8.42 (Optional extension: three infinite-dimensional hypotheses). Three notions must not be conflated. First, a Hilbert space is an ambient inner-product space complete for its norm; the finite-support space c_{00} is not complete. Second, an orthonormal set is *complete* when its closed span is the whole Hilbert space; $\{e_1\}$ is orthonormal but not complete in ℓ^2, so Bessel is strict at e_2. Third, a subspace must be *closed* for the Hilbert-space decomposition $H = M \oplus M^\perp$; c_{00} is a dense non-closed subspace of ℓ^2 with $c_{00}^\perp = \{0\}$. Complete ON sets give expansion/Parseval, and closed subspaces give projection/decomposition. Lecture 13 develops these separate infinite-dimensional theorems.

The remaining ingredient is the adjoint. We have seen (Riesz) that inner products turn vectors into functionals; next lecture we turn *operators* into their adjoints $T^\dagger$ via $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$, and single out the self-adjoint, unitary, and normal operators—the ones whose eigenvectors form an orthonormal basis, by the spectral theorem of Lecture 10.

Summary of main concepts

Here V is finite-dimensional over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and every basis sum is finite. ONB coordinates, Parseval, identity resolution, and Gram–Schmidt are finite formulas. For $U \subseteq V$, projection gives $V = U \oplus U^\perp$, best approximation, and Bessel. Finite Riesz gives each $\varphi : V \to \mathbb{F}$ a unique representative, with the field-dependent ket-to-bra behavior.

- **Finite orthonormal basis**—finite coordinate, Parseval, and identity-resolution sums.
- **Gram–Schmidt**—the finite ONB algorithm, preserving partial spans.
- **Finite projection**—$V = U \oplus U^\perp$, best approximation, and Bessel.
- **Finite Riesz**—unique $\varphi = \langle w, \cdot\rangle$; ket-to-bra is conjugate-linear over $\mathbb{C}$ and linear over $\mathbb{R}$.

Exercises for this lecture are in the companion sheet `exercises-8.pdf`.